Recovered drift and diffusion models

SSISDE experiment results

20 October 2026

Final Stochastic SINDy fits using noise recovered from each drift method: $dX_t = \widehat{\mu}(X_t)\,dt + \widehat{\sigma}(X_t)\,dW_t$.

Method	Recovered drift $\widehat{\mu}(x)$	Recovered diffusion $\widehat{\sigma}(x)$	α	ρ
Black–Scholes				
Method 1	$0.5038\,x$	$0.0032 + 0.2958\,x$	0.0004	0.8500
Method 2	$0.5017\,x$	$0.0032 + 0.2958\,x$	0.0004	0.8500
ODE2	$0.5019\,x$	$0.0032 + 0.2958\,x$	0.0004	0.8500
Constant coefficients				
Method 1	$0.4997 + 0.0003\,x$	1.0002	0.0001	0.7000
Method 2	0.5000	1.0002	0.0006	0.8500
ODE2	0.5000	1.0002	0.0006	0.8500
Quadratic coefficients				
Method 1	$0.6205\,x + 0.3590\,x^2$	$-0.0178 + 0.5360\,x + 0.0842\,x^2$	0.0009	0.5000
Method 2	$1.0221\,x$	$-0.0200 + 0.5401\,x + 0.0824\,x^2$	0.0009	0.4000
ODE2	$0.4273\,x + 0.4510\,x^2$	$-0.0202 + 0.5402\,x + 0.0825\,x^2$	0.0009	0.4000
Ornstein–Uhlenbeck				
Method 1	$-0.5003 + 0.0003\,x$	0.2990	0.0009	0.4000
Method 2	-0.5000	0.2990	0.0009	0.4000
ODE2	$-0.5000\,x$	0.2990	0.0002	0.4000

Parameters. α is the ElasticNet regularization strength; ρ is its L1 mixing ratio. Each pair applies to the joint drift–diffusion fit and was selected by seven-fold cross-validation with the one-standard-error rule. All four experiments used 100,000 simulated samples.

Selection. The double-well outer search over polynomial degrees and regression types has been removed. The constant experiment uses its original ρ grid: 0.7000, 0.8500, 0.9500, 1.0000.

Precision. All displayed coefficients, α, and ρ are rounded to four decimal places. Terms that round to zero are omitted from the displayed polynomials. Full-precision coefficients and parameters remain in the result files; rounding does not change the fitted models. $\widehat{\sigma}$ is the diffusion amplitude, not its square.

Source. `outputs/<case>/ssisde_comparison.json`, where `<case>` is `black_scholes`, `constant`, `quadratic`, or `ou`.